Homework 3

Liv Zou

2026-05-26

Table of contents

Setup

```
library(tidyverse)
library(here)
library(janitor)
library(readxl)
```

Loading kelp package:

```
kelp <- read_csv(here::here("..", "data", "temp-kelp (1).csv"))
# Display first 6 rows
head(kelp)
```

```
# A tibble: 6 x 2
  temp_c kelp_elong
```

```
   <dbl>      <dbl>
1   19.2        6.5
2   16.3        6
3   19.1        5.8
4   17.2        6.8
5   17.3        6.9
6   18.1        7.1
```

Problem 1

1a.

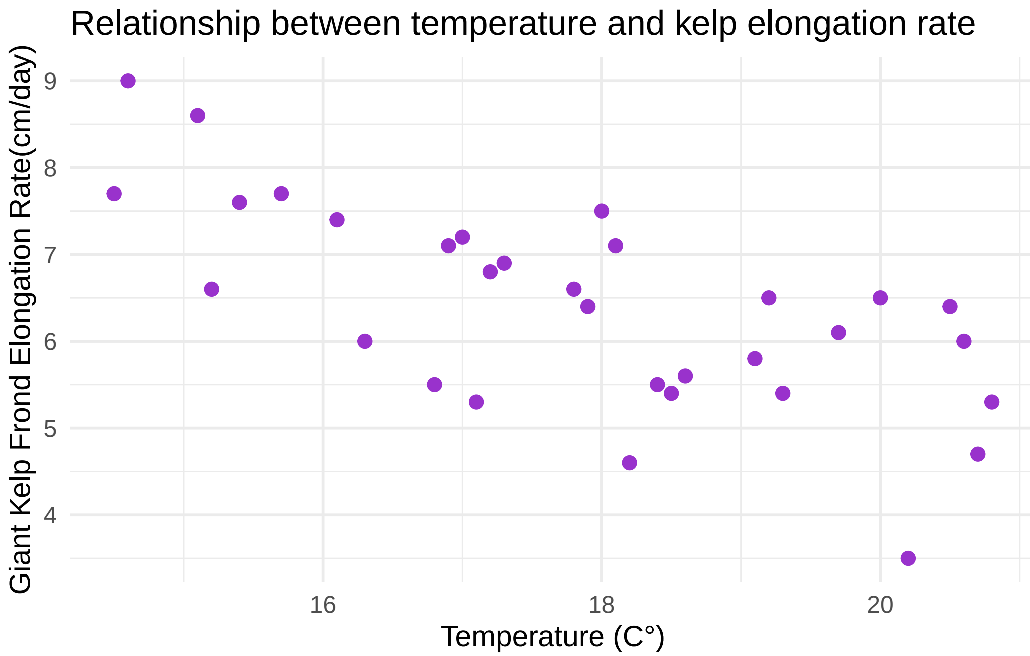
The appropriate tests to determine the strength of the relationship between temperature and giant kelp frond elongation rate are the Pearson correlation test and the Spearmann rank correlation test.

The Pearson correlation test is for parametric data and assumes a linear relationship between continuous variables, approximate normal distribution, and absence of outliers. The Spearman rank correlation test is a non-parametric test that does not require the data to be normally distributed, and is appropriate when the data is not linear or when assumptions from the Pearson test have been violated.

Both tests are similar because they both measure strength and direction of a relationship between two variables and don't assume causation in either test. However, both tests are also different. The Spearmann test uses a ranking of the data than raw values, and isn't affected by outliers or requiring the data to have normality. It is best used in monotonic relationships, where the direction does not change. The Pearson test uses raw data values from the dataset and measures the strength of linear relationships in continuous variables. It assumes normality and absence of outliers.

1b.

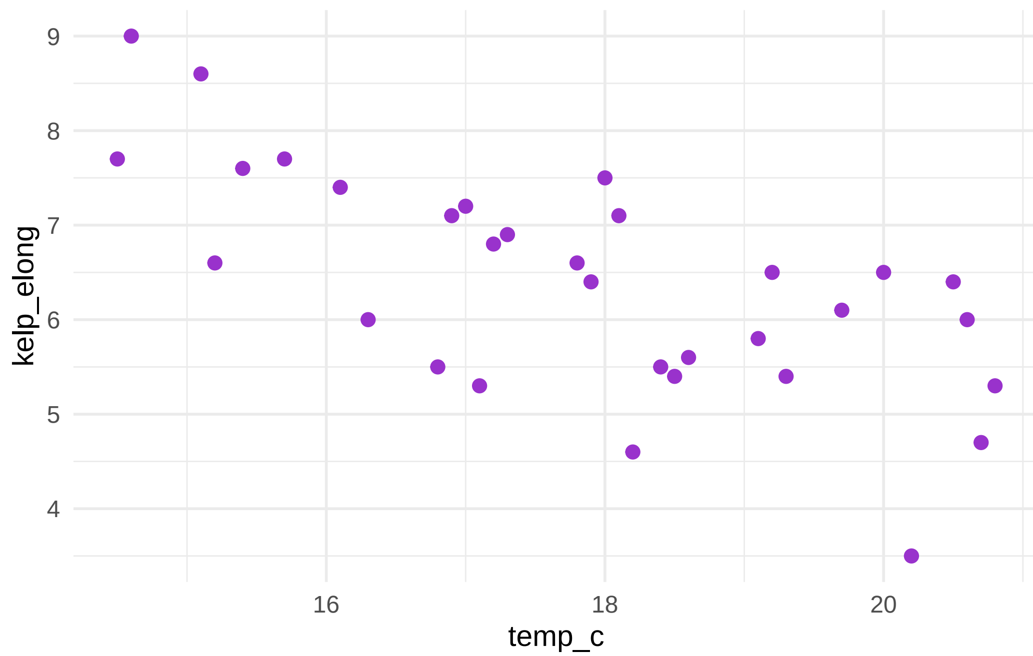
```r
# Set up scatterplot with ggplot
ggplot(kelp, aes(x = temp_c, y = kelp_elong)) +
   # Set individual observations
  geom_point(color = "darkorchid", size = 2) +
  labs(
    # Set up x y axes for scatterplot
    x = "Temperature (C°)",
       y = "Giant Kelp Frond Elongation Rate(cm/day)",
       title = "Relationship between temperature and kelp elongation rate") +
theme_minimal()
```



1c.

Assumption check:

```
# Pasting data from scatterplot
ggplot(kelp, aes(x = temp_c, y = kelp_elong)) +
  geom_point(color = "darkorchid", size = 2) +
  theme_minimal()
```



Pearson correlation:

```r
# Using pearson correlation
cor.test(
  kelp$temp_c,
  kelp$kelp_elong,
  method = "pearson"
)
```

```
    Pearson's product-moment correlation

data:  kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
       cor
-0.6877489
```

From my data, I was able to examine information from my scatterplot, and used it for checking my assumptions for using a Pearson correlation test. I assessed if the relationship between temperature and growth rate appeared linear and if there were any glaring outliers, to which it did not and the data appeared linear, having a generally negative trend. Therefore, a Pearson's test was most appropriate to use.

1d.

A Pearson's correlation test found a significant negative relationship between ocean temperature (C°) and giant kelp frond elongation rate ($r = -0.688$, $p < 0.001$), indicating that as ocean temperature increases the growth rate tends to decrease. A p-value less than 0.001 suggests statistical significance, and the correlation coefficient provided suggests a moderate to strong negative association.

1e.

Our results suggest rising ocean temperatures negatively impact giant kelp frond growth rates, which could negatively influence surrounding ecosystems and affect biodiversity. It raises a biological and ecological concern for surrounding members in aquatic ecosystems that coexist around these kelp populations as kelp is an abundant autotroph, habitat, and food source for many marine organisms.

Reduced growth would massively influence other members in the associated food chains/ecosystems and could deteriorate ecosystem functioning. These statistical findings call for an action to further monitor on kelp communities, as continuing increases in ocean temperatures may pose significant danger and risk to ecosystems, climate, nature, and our environment.

These concerns raised from these statistical findings should not be taken lightly and should be used towards greater ecological preservation efforts and continued environmental management.

1f.

```r
# pearson test
cor.test(kelp$temp_c, kelp$kelp_elong, method = "pearson")
```

```

	Pearson's product-moment correlation

data:  kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
       cor 
-0.6877489 
```

```
# Spearmann test
cor.test(
  kelp$temp_c,
  kelp$kelp_elong,
  method = "spearman",
# Suppress warnings of duplicate values in data set
  exact = FALSE
)
```

```

	Spearman's rank correlation rho

data:  kelp$temp_c and kelp$kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho 
-0.6891684 
```

The Spearmann rank correlation test found a significant negative relationship (= -0.689, p-value < 0.001), similar to the results of the Pearson correlation test ($r = -0.688$, $p < 0.001$). Since both tests had nearly identical correlation coefficients and small p-values, it would still have lead to the same decision of rejecting the null hypothesis of no relationship between the variables. These results provide supporting evidence of the trend that higher ocean temperatures are associated with lower kelp growth rates.

Problem 2

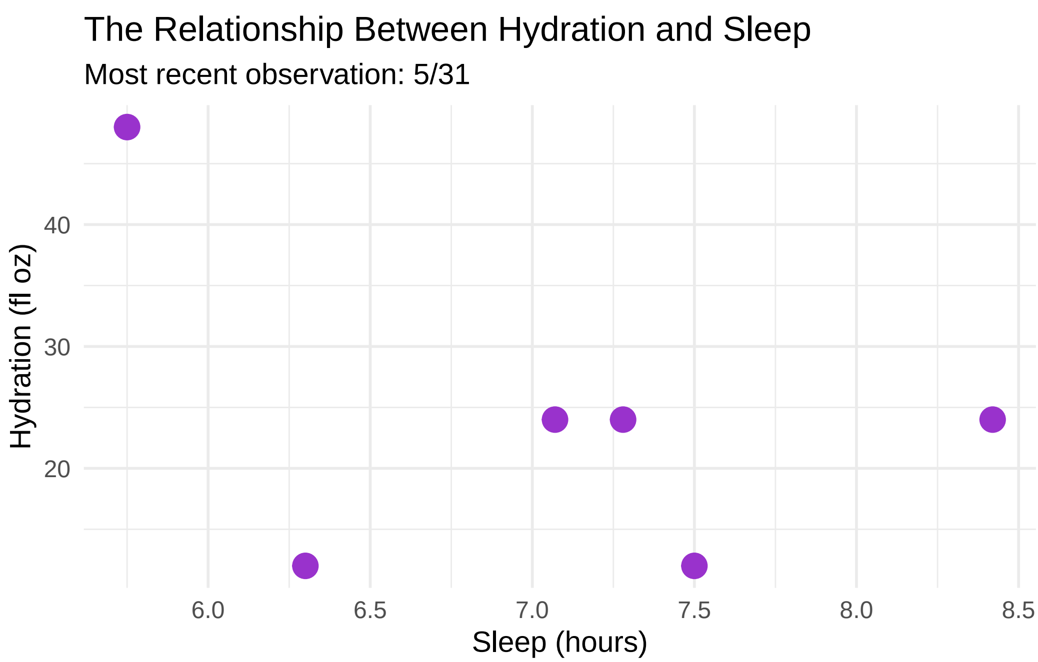
2a.

```
# defining csv file into R
personal_data <- read_csv("homework 3 personal data project - Sheet1.csv")
# Converting into date format in Rstudio
personal_data$Date
```

```
[1] "5/26" "5/27" "5/28" "5/29" "5/30" "5/31"
```

Most recent data:

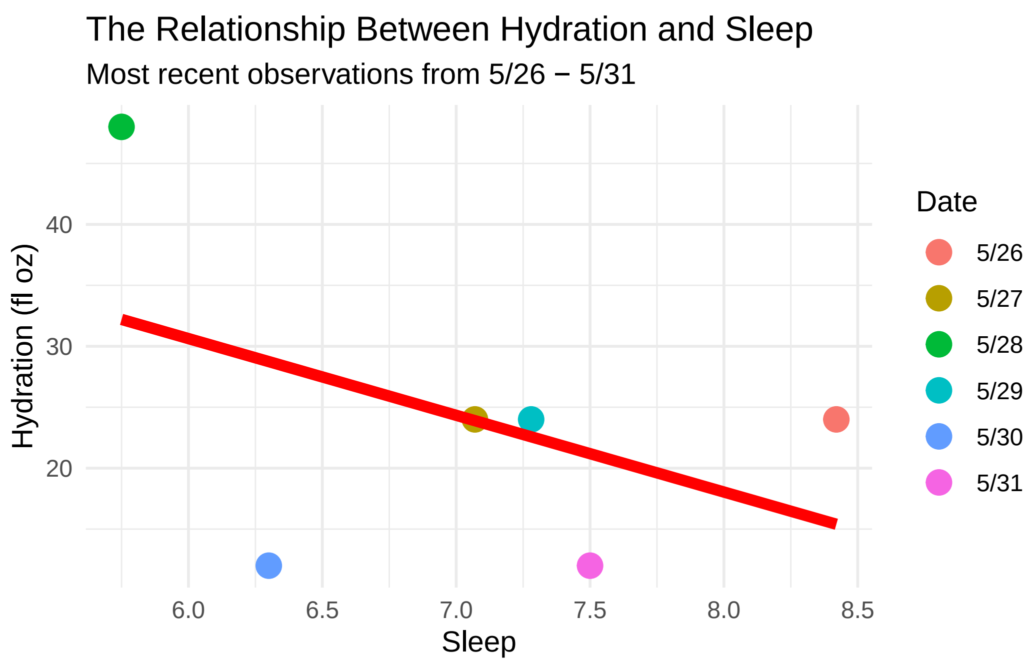
```r
most_recent_date <- max(personal_data$Date, na.rm = TRUE)

ggplot(personal_data,
       aes(x = Sleep_hrs,
           y = hydration)) +
  geom_point(color = "darkorchid", size = 4) +
  labs(
    x = "Sleep (hours)",
    y = "Hydration (fl oz)",
    title = "The Relationship Between Hydration and Sleep",
    subtitle = paste("Most recent observation:", most_recent_date)
  ) +
  scale_y_continuous(labels = scales::comma) +
  theme_minimal()
```



Hydration + Sleep:

```r
# Set up scatterplot
ggplot(personal_data,
       aes(y = hydration,
           x = Sleep_hrs,
           color = Date)) +
  geom_point(size = 4) +
```

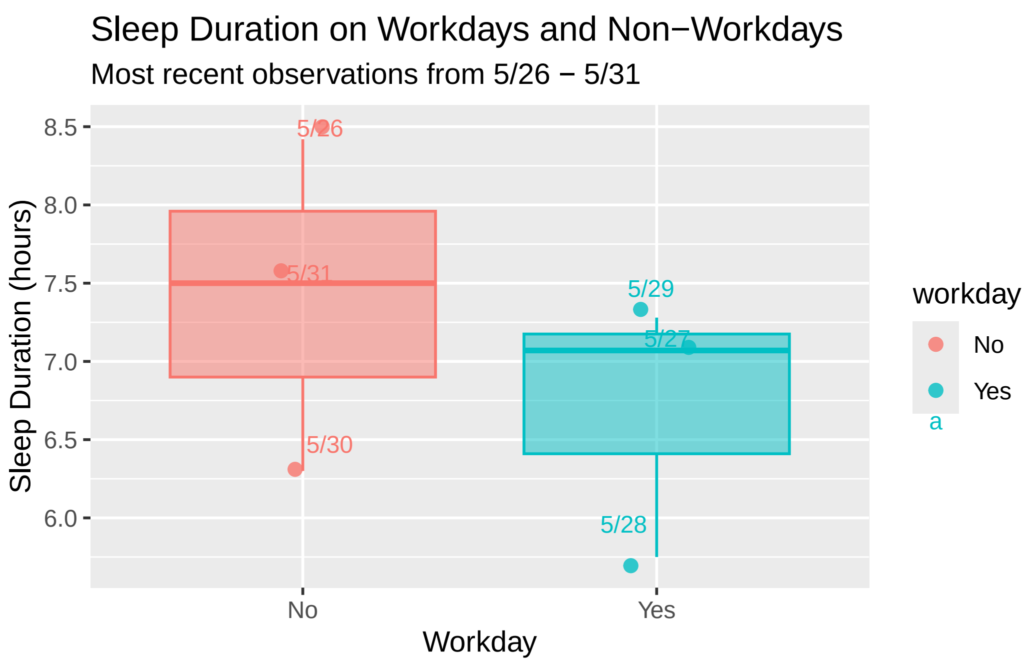
```r
# Add linear regression line
  geom_smooth(method = "lm",
              se = FALSE,
              color = "red",
              linewidth = 2) +
  labs(
    # Labelling x y axes
    y = "Hydration (fl oz)",
    x = "Sleep",
    color = "Date",
    title = "The Relationship Between Hydration and Sleep",
    subtitle = "Most recent observations from 5/26 - 5/31"
  ) +
  # Format y-axis labels
  scale_y_continuous(labels = scales::comma) +
  theme_minimal()
```



Workday vs. Sleep

```r
# Defining points and dataset in ggplot
ggplot(personal_data, aes(x = workday, y = Sleep_hrs, fill = workday, color = workday)) +
  # Making a boxplot to show distribution of sleep on workdays + nonworkdays
  geom_boxplot(show.legend = FALSE, alpha = 0.5) +
```

```r
  geom_jitter(width = 0.1, alpha = 0.8, size = 2) +
  # Manually assigning specific points to specific individual dates
  geom_text(
    aes(label = Date),
    position = position_jitter(width = 0.1),
    vjust = -0.8,
    size = 3
  ) +
  # Set up x y axes
  labs(
    x = "Workday",
    y = "Sleep Duration (hours)",
    title = "Sleep Duration on Workdays and Non-Workdays",
    subtitle = "Most recent observations from 5/26 - 5/31")
```



2b.

Lineplot: Figure 1. Nightly sleep (in hrs and minutes) and daily hydration levels (in fluid ounces) from 5/26/2026-5/31/2026. Each point and color represents a different, individual day of observation. The red line in the graph shows the general trend of the relationship between sleep and hydration. The negative trend observed in this graph suggests that increased sleep is generally associated with lower levels of hydration. However, many of the points are scattered, and the data does not seem to be linear.

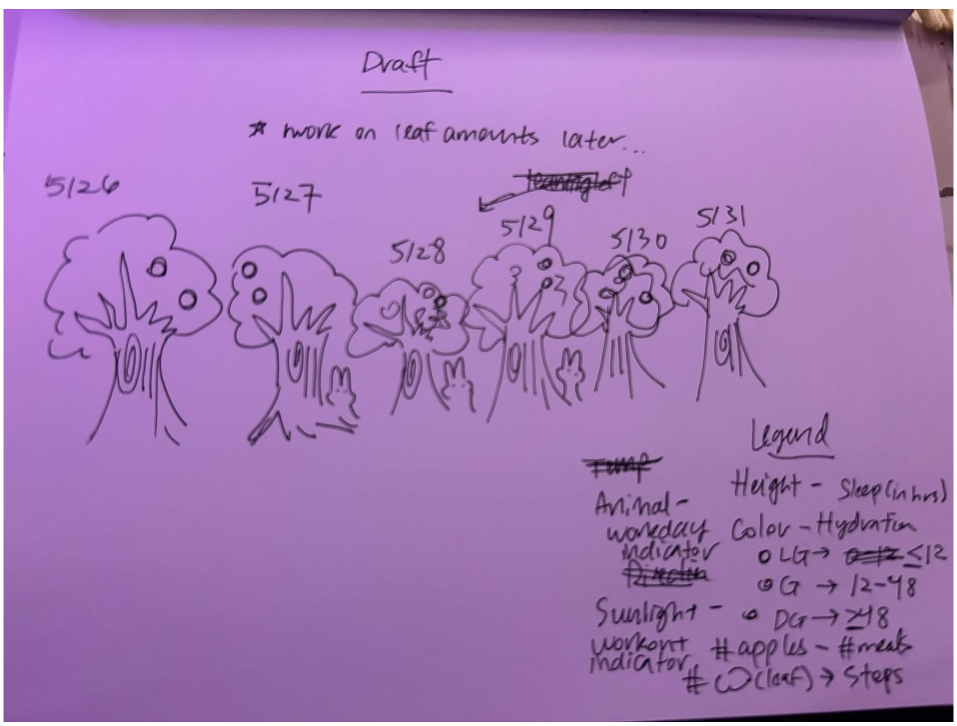
Boxplot: Figure 1. Nightly sleep duration (in hours) on workdays and non-workdays from 5/26/2026-5/31/2026. Each point represents an individual observation, with yes points indicated with blue and no points indicated in red. The boxplots summarize the overall distribution of sleep duration depending on whether there was a workday or not. Sleep duration tended to be generally higher if the day was not a work day.

Problem 3

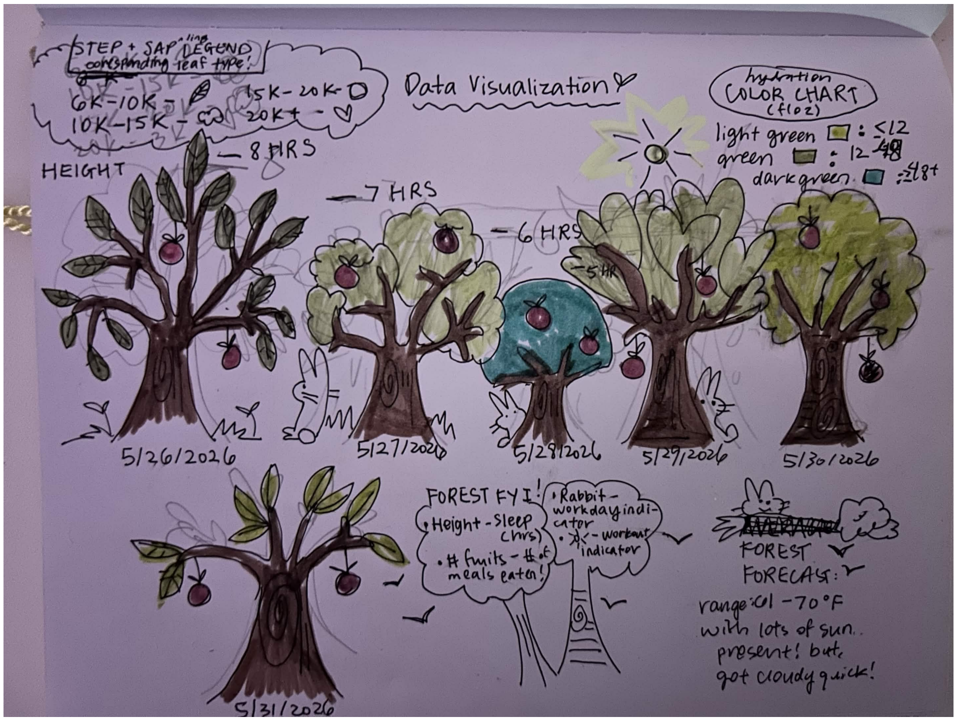
3a.

I think an affective visualization that could be useful for my project would be in the form of a tree, where the more sleep there is, the taller the tree is. The color of the leaves in the tree could represent hydration intake, with light to dark green showing smaller to larger hydration levels. The amount of leaves could represent how many steps there are. Meals eaten could be the number of fruit present. Workout days could be represented by the presence of a sun. Workdays could be represented with the presence of animals.

3b.



3c.



3d.

My data visualization was drawn by hand on paper and it is showing health characteristics such as sleep, hydration, meal intake, in the allegory of a forest of trees. I was particularly interested of the illustration of the glacial oil rigs shown during class, as well as looking through Jill Pelto's work. I used mildliner zebra highlighters to color my work and the sarasa 0.5 ink pen (the same one I use to journal) to write and outline, and I sketched everything first in pencil.

3e.

Link to slides:

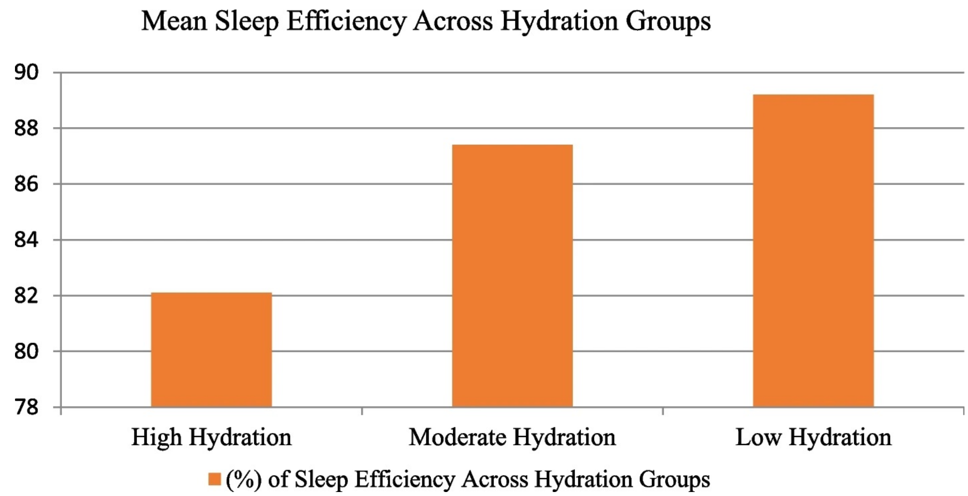
Google Slides Presentation

Problem 4

4a.

Paper for reference: https://doi.org/10.1186/s41606-025-00169-0

In this paper, the authors wanted to see how hydration habits before sleeping affected the quality of sleep in adults. They used a one-way ANOVA test to compare sleep quality across three groups distributed by hydration levels: high, moderate, low. For post-hoc, they used the Tukey's HSD with Bonferroni correction to assess differences. The response variable(s) is sleep quality, efficiency and no. of nocturnal awakenings, and the predictor variable is hydration levels ranked from high to low.



4b.

The authors were very clear with their statistics in this figure, because they categorized their data about the relationship between sleep quality and hydration levels (in orange) through an ordinal bar graph. Here, I can clearly see that the lower hydration group produces a generally high mean sleep efficiency, while the higher hydration group produces a generally lower sleep efficiency. They do not include the error bars, CI, or participant data, which makes it difficult to see the actual spread of data at its capacity, and the variability within each group. What is interesting is that in the paper, they say that there are error bars and standard deviations and it is actually the Y-axis, which is difficult to interpret, and they assess sleep efficiency as a percentage in combination of the orange bars.

4c.

I think they have a pretty high data::ink ratio because the graph is really simple and minimalistic, and there's more ink dedicated to the actual display/representation of their statistical summaries and results than for anything else. They handled the visual clutter really well, because the design is minimal and there is only three categories, focusing primarily on the mean sleep efficiency percentage across groups.

4d.

I think the authors should add error bars and individual data points from a small sample of each category would be helpful because it would help readers see how spread out the actual data is rather than just through a summarization. I also think the lines in the background could be removed, and a caption addressing the figure in more depth could be helpful for readers to interpret more easily.